

Assignment 2 : Final Report

Dataset Description and Preprocessing

The Global Pollution Analysis dataset contains 200 records with environmental and economic indicators such as pollution indices, industrial waste, CO₂ emissions, renewable energy, population, GDP per capita, and energy recovery.

During preprocessing, the dataset was checked for missing values, and none were found. The categorical columns **Country** and **Year** were encoded using Label Encoding, while the pollution indices were standardized using **StandardScaler**. After preprocessing, exploratory data analysis was performed to better understand the dataset.

Exploratory Data Analysis

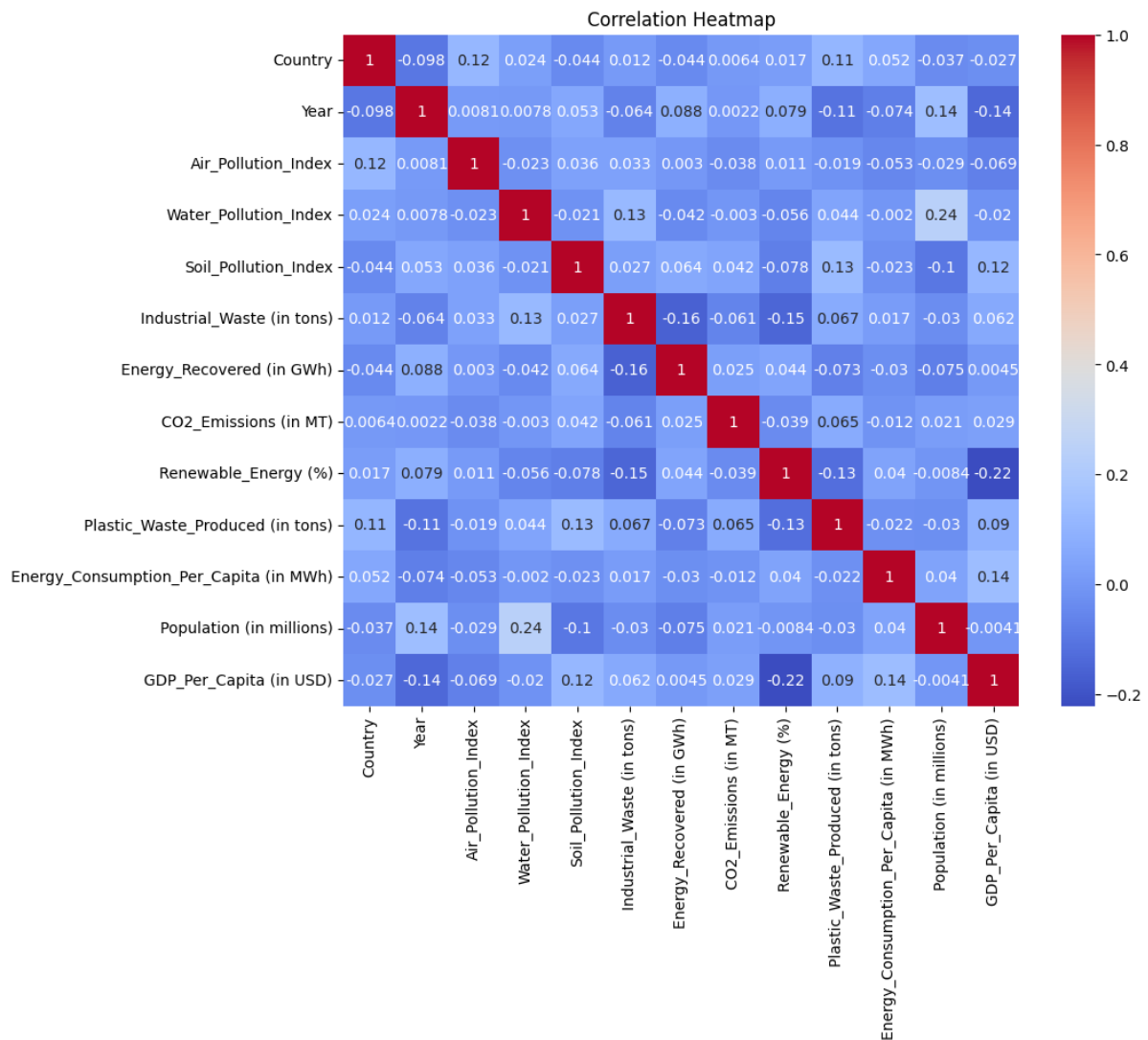
The analysis showed that **Air Pollution** was generally higher than Water and Soil Pollution. Correlation analysis revealed weak relationships between **Energy Recovery** and most environmental factors. Visualizations also showed that pollution levels fluctuated over time rather than following a consistent increasing or decreasing trend. Additionally, Air Pollution displayed the greatest variation among the three pollution indices.

Feature Engineering

A Pollution Score was created by averaging the Air, Water, and Soil Pollution Indices. Yearly trend analysis was performed to understand changes in pollution and energy recovery over time. The dataset already contained the Energy Consumption Per Capita feature, which was used directly for analysis.

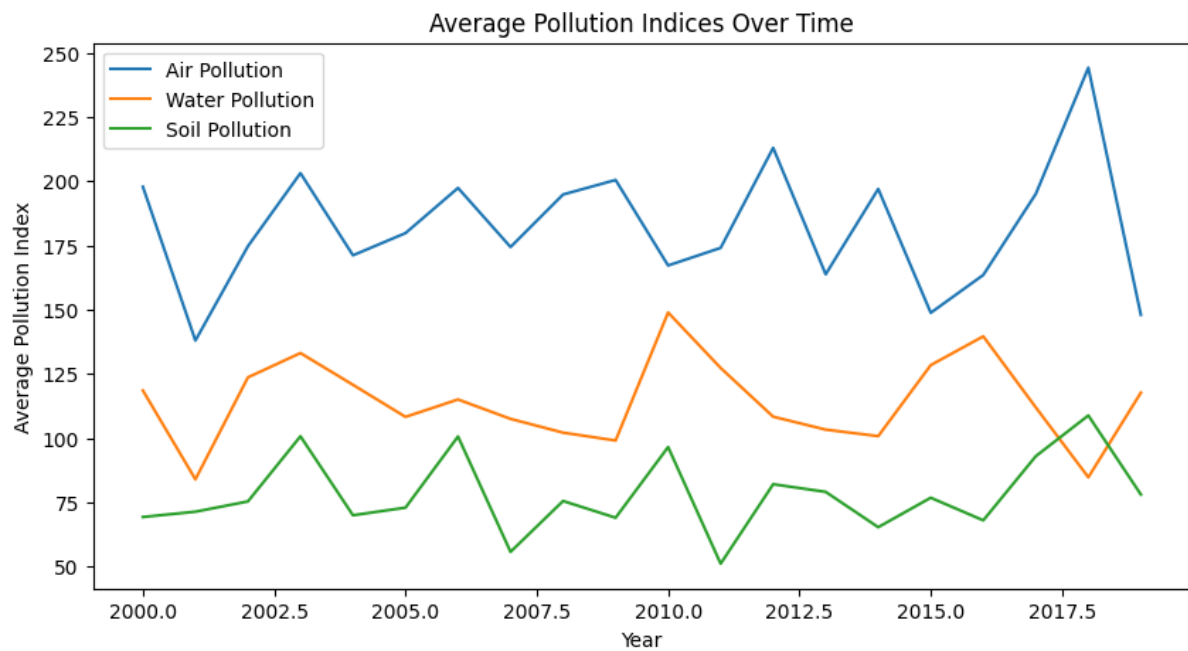
Data Visualization:

1. Correlation Heatmap



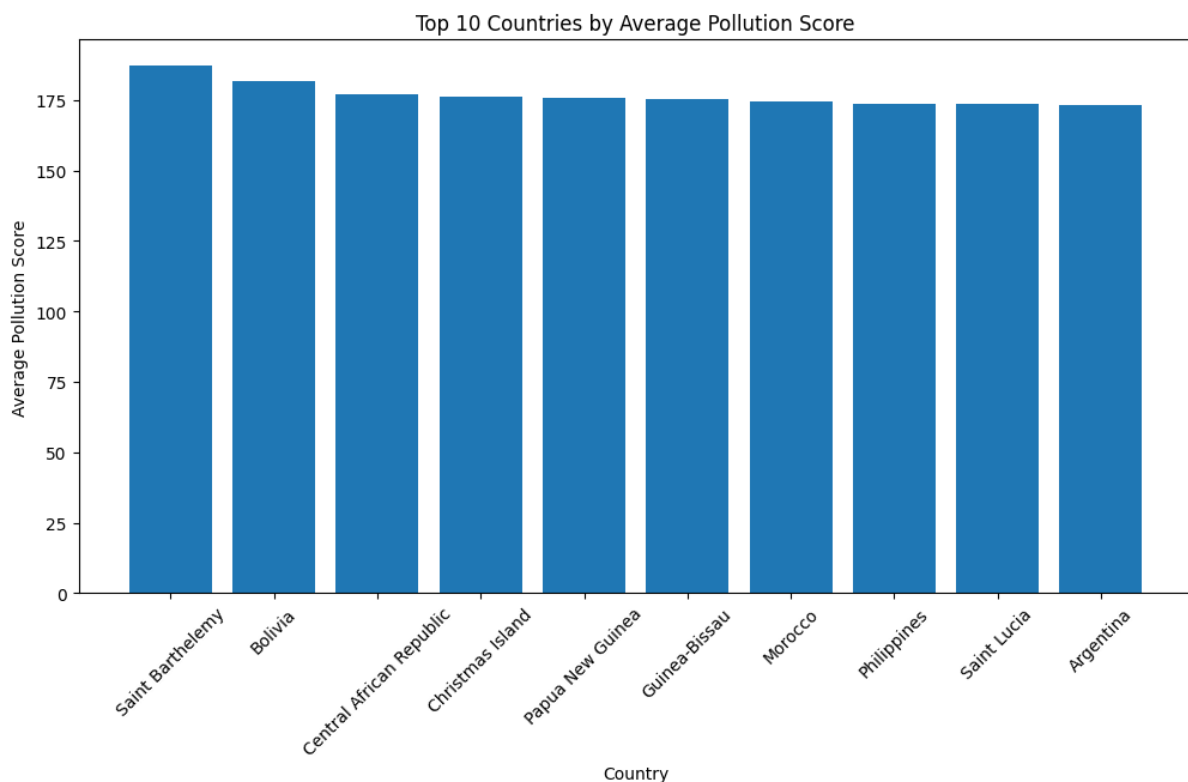
The heatmap was used to analyze relationships between numerical variables. Most features showed weak correlations with Energy Recovery, indicating limited linear relationships in the dataset.

2. Pollution Trends Over Time (Line Plot)



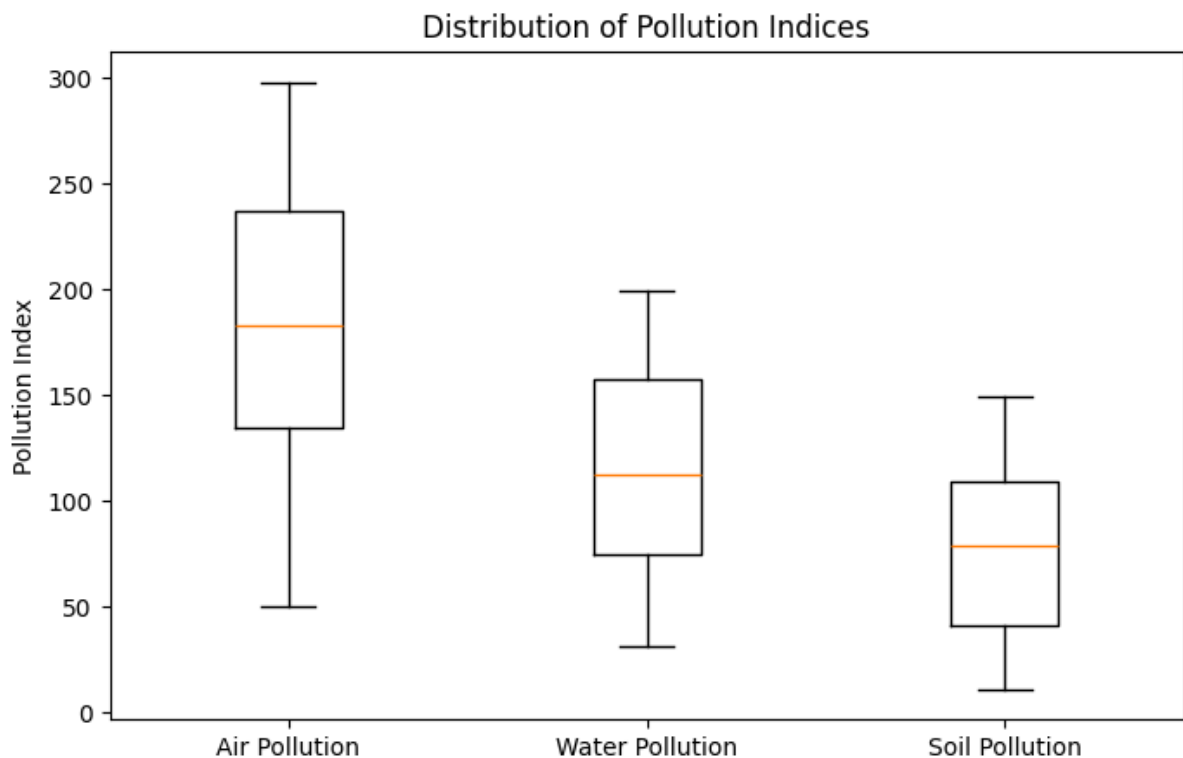
The line plot illustrates the average Air, Water, and Soil Pollution Indices over time. Air Pollution remained consistently higher than Water and Soil Pollution, while all three pollution types fluctuated across years.

3. Top 10 Countries by Average Pollution Score (Bar Chart)



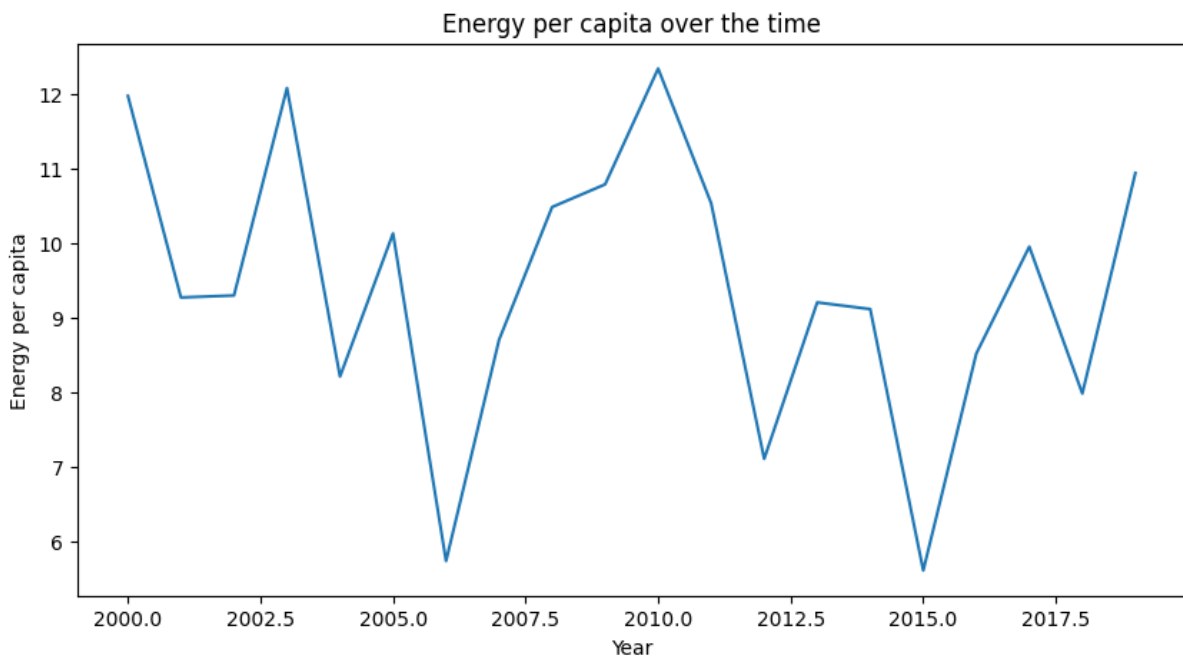
The bar chart compares the top 10 countries with the highest average pollution scores. Saint Barthelemy recorded the highest pollution score among the countries analyzed.

4. Distribution of Pollution Indices (Box Plot)



The box plot compares the distribution of Air, Water, and Soil Pollution Indices. Air Pollution exhibited the highest median and the greatest variability, while no significant outliers were observed.

5. Energy Consumption Per Capita Trend



The visualization shows how average energy consumption per capita changed over time. The values fluctuated across years without a clear increasing or decreasing trend.

- **Linear Regression Model:**

A Linear Regression model was developed to predict Energy Recovery (in GWh) using environmental indicators such as Air Pollution Index, Soil Pollution Index, Industrial Waste, and CO₂ Emissions.

Results

- R² Score: -0.015
- Mean Squared Error (MSE): 24539.71
- Mean Absolute Error (MAE): 141.35

Interpretation

The negative R² score indicates that the model was unable to effectively explain the variation in Energy Recovery. The high MAE and MSE values further suggest that the predictions deviated significantly from the actual values. This performance is likely due to the weak correlations observed during EDA.

- **Logistic Regression Model**

A Logistic Regression model was developed to classify countries into Low, Medium, and High Pollution categories.

Results

- Accuracy: 42.5%

Interpretation

The model performed slightly better than random guessing for a three-class classification problem. It was most effective at identifying Low Pollution categories, while Medium Pollution categories were more difficult to classify accurately.

Model Comparison

Model	Objective	Performance
Linear Regression	Predict Energy Recovery	R ² = -0.015
Logistic Regression	Classify Pollution Level	Accuracy = 42.5%

The Logistic Regression model performed better relative to its objective, whereas the Linear Regression model struggled due to weak relationships between the target variable and the selected features.

Key Insights and Recommendations

- Air Pollution levels were generally higher than Water and Soil Pollution levels.
- Pollution levels fluctuated over time without a clear long-term trend.
- Energy Recovery showed weak correlations with most environmental indicators.

- Logistic Regression achieved moderate classification performance, while Linear Regression showed poor predictive capability.

Recommendations

1. Collect additional environmental and industrial variables that may influence Energy Recovery.
2. Apply advanced machine learning algorithms such as Random Forest or Gradient Boosting to improve performance.
3. Perform further feature engineering to capture non-linear relationships within the dataset.
4. Increase dataset size to improve model generalization and predictive accuracy.